CardDemo interest calculation: wiki pages and I/O contract

Wiki pages generated by the FlowX.AI legacy transformation agent from AWS CardDemo (open source, Apache-2.0). CONTRACT.md is the input/output contract for the build.

Pages included:

- `02-processes/process-interest-calculation.md`
- `10-acceptance-criteria/acceptance-criteria-interest-calculation.md`
- `01-data-model/project-data-model.md`
- `CONTRACT.md`

`02-processes/process-interest-calculation.md`

Process: Interest Calculation (Batch Pipeline)

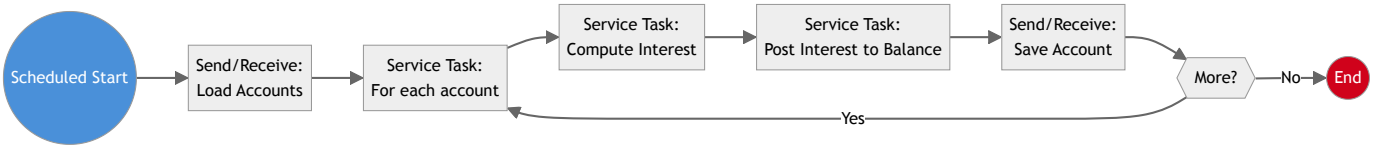
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Field	Value
Name	Interest Calculation
Type	Scheduled Process (orchestration)
Legacy Source	`CBACT04C` (+ `CBACT01–03C`) via `INTCALC.jcl`; disclosure config `DISCGRP.jcl`
Status	Draft
Related Data	Account, TransactionCategoryBalance

Business Description

Periodic pipeline that computes interest for each account based on its balance and disclosure/interest group, then posts the interest to the account. Replaces the `CBACT04C`/`INTCALC` batch. The interest-rate parameters come from the disclosure-group reference data (`DISCGRP`).

Process Flow



Nodes

#	Node	Type	Description	Legacy Reference
1	Scheduled Start	Start Event (Scheduled Trigger)	Interest cycle	`INTCALC.jcl`
2	Load Accounts	Send/Receive Message Task	Read accounts	`CBACT01C` read
3	For each account	Service Task (loop)	Iterate	main loop
4	Compute Interest	Service Task (NoOpHandler)	`interest = balance × rate(group)`	`CBACT04C`
5	Post Interest	Service Task (NoOpHandler)	Add interest to balance	balance update
6	Save Account	Send/Receive Message Task	Persist	`REWRITE ACCTFILE`

Business Rules

Rule: ComputeInterest

- **Legacy source:** `CBACT04C` interest computation using disclosure-group rate.
- **Logic:** `interest = currentBalance × interestRate(groupId)`; posted to balance and category balance.
- **FLOWX implementation:** NoOpHandler; interest rate from disclosure-group reference data / Configuration Parameter.
- `{{TODO: Extract exact interest formula, rounding, and grouping from CBACT04C + DISCGRP.}}`

Integration Touchpoints

Send Node	Topic	Payload	Receive	System
Load Accounts	`ai.flowx.in.carddemo.datastore.loadaccounts`	`{ page }`	Account[]	FlowX Persistence
Save Account	`ai.flowx.in.carddemo.datastore.saveaccount`	Account	ack	FlowX Persistence

Swimlanes

- **System**

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Acceptance Criteria: Interest Calculation

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Verifies Interest Calculation (`CBACT04C`/INTCALC).

Configuration Dependencies

Config	Effect
Disclosure-group rates (`DISCGRP`)	Interest rate per account group

Scenarios

#	Scenario	Steps	Expected Result
1	Compute interest (happy)	Run over accounts	Interest = balance × group rate; matches legacy
2	Rounding (boundary)	Balance causing fractional cents	Rounding matches legacy `{{TODO: confirm rounding mode}}`
3	Zero balance (edge)	Account balance 0	Zero interest
4	Multiple disclosure groups (gateway)	Accounts in different groups	Correct rate per group

Data Mapping Verification

Flow	Verify
Compute Interest	balance × rate(groupId) → posted interest

Legacy Comparison

Input	Legacy `CBACT04C`	FLOWX
account set	interest amounts	identical per account

Project Data Model — AWS CardDemo → FLOWX.AI

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This section documents all **Project Data Types** derived from the CardDemo copybooks (`/app/cpy`). Each VSAM record layout maps to a Project Data Type; COBOL `05`-level items become attributes. Packed/zoned/signed numerics (`COMP-3`, `S9(n)V99`) map to `NUMBER`, fixed-width `X(10)` dates map to `DATE`, and FILLER tails are dropped. Enumerated reference codes are documented in Enumerations.

> **Type-conversion note:** `S9(10)V99` → `NUMBER` (decimal, scale 2); `9(n)` → `NUMBER` (integer); `X(n)` → `STRING` (maxLength n); `X(10)` date → `DATE`; `X(26)` timestamp → `DATE` (timestamp). The `ACCT-EXPIRAION-DATE` legacy typo is corrected to `expirationDate` — see `{{DECISION}}` below.

Business Categories

Category	Types
Core Processing	Account, Transaction, TransactionCategoryBalance
Domain	Customer, Card, CardXref
Reference / Config	TransactionType, TransactionCategory
Security	SecurityUser
Navigation (internal)	SessionContext

Account

Replaces the ACCOUNT-RECORD VSAM KSDS (`ACCTFILE` / `CVACT01Y`, 300 bytes). Key: `accountId`.

Attribute	Type	Required	Constraints	Description	Legacy Source
accountId	NUMBER	Yes	11 digits, unique key	Account identifier	`ACCT-ID 9(11)`
activeStatus	ENUM	Yes	AccountStatus	Active status flag	`ACCT-ACTIVE-STATUS X(1)`
currentBalance	NUMBER	Yes	decimal(12,2), signed	Current account balance	`ACCT-CURR-BAL S9(10)V99`
creditLimit	NUMBER	Yes	decimal(12,2), signed	Credit limit	`ACCT-CREDIT-LIMIT S9(10)V99`
cashCreditLimit	NUMBER	Yes	decimal(12,2), signed	Cash credit limit	`ACCT-CASH-CREDIT-LIMIT S9(10)V99`
openDate	DATE	Yes		Account open date	`ACCT-OPEN-DATE X(10)`
expirationDate	DATE	Yes		Account expiration date	`ACCT-EXPIRAION-DATE X(10)` *(legacy typo)*
reissueDate	DATE	No		Account reissue date	`ACCT-REISSUE-DATE X(10)`

Attribute	Type	Required	Constraints	Description	Legacy Source
currentCycleCredit	NUMBER	Yes	decimal(12,2), signed	Current-cycle credits	`ACCT-CURR-CYC-CREDIT S9(10)V99`
currentCycleDebit	NUMBER	Yes	decimal(12,2), signed	Current-cycle debits	`ACCT-CURR-CYC-DEBIT S9(10)V99`
addressZip	STRING	No	max 10	Account ZIP code	`ACCT-ADDR-ZIP X(10)`
groupId	STRING	No	max 10	Disclosure/interest group id	`ACCT-GROUP-ID X(10)`

`{{DECISION: Correct the legacy field name typo ACCT-EXPIRAION-DATE → expirationDate (recommended), or preserve the misspelling for round-trip fidelity? ETL must map the legacy column either way.}}`

Used by: Account View, Account Update, Bill Payment, Transaction Posting, Interest Calculation.

Customer

Replaces CUSTOMER-RECORD (`CUSTFILE` / `CVCUS01Y`, 500 bytes). Key: `customerId`.

Attribute	Type	Required	Constraints	Description	Legacy Source
customerId	NUMBER	Yes	9 digits, unique key	Customer identifier	`CUST-ID 9(9)`
firstName	STRING	Yes	max 25	First name	`CUST-FIRST-NAME X(25)`
middleName	STRING	No	max 25	Middle name	`CUST-MIDDLE-NAME X(25)`
lastName	STRING	Yes	max 25	Last name	`CUST-LAST-NAME X(25)`
addressLine1	STRING	Yes	max 50	Address line 1	`CUST-ADDR-LINE-1 X(50)`
addressLine2	STRING	No	max 50	Address line 2	`CUST-ADDR-LINE-2 X(50)`
addressLine3	STRING	No	max 50	Address line 3	`CUST-ADDR-LINE-3 X(50)`
addressStateCode	STRING	No	max 2	State code	`CUST-ADDR-STATE-CD X(2)`
addressCountryCode	STRING	No	max 3	Country code	`CUST-ADDR-COUNTRY-CD X(3)`
addressZip	STRING	No	max 10	ZIP code	`CUST-ADDR-ZIP X(10)`
phoneNumber1	STRING	No	max 15	Primary phone	`CUST-PHONE-NUM-1 X(15)`
phoneNumber2	STRING	No	max 15	Secondary phone	`CUST-PHONE-NUM-2 X(15)`
ssn	NUMBER	Yes	9 digits (PII)	Social security number	`CUST-SSN 9(9)`
govtIssuedId	STRING	No	max 20	Government-issued id	`CUST-GOVT-ISSUED-ID X(20)`
dateOfBirth	DATE	No		Date of birth	`CUST-DOB-YYYY-MM-DD X(10)`
eftAccountId	STRING	No	max 10	EFT account id	`CUST-EFT-ACCOUNT-ID X(10)`
primaryCardHolderInd	STRING	No	max 1	Primary-holder indicator	`CUST-PRI-CARD-HOLDER-IND X(1)`
ficoCreditScore	NUMBER	No	300–850	FICO credit score	`CUST-FICO-CREDIT-SCORE 9(3)`

{{TODO: SSN and govt id are PII — define masking/encryption and access restrictions in FLOWX.}}

Used by: Account View, Account Update, Authorization.

Card

Replaces CARD-RECORD (CARDFILE + CARDAIX / CVACT02Y, 150 bytes). Key: cardNumber; alternate index on cardAccountId.

Attribute	Type	Required	Constraints	Description	Legacy Source
cardNumber	STRING	Yes	16 chars, unique key (PCI)	Card number	CARD-NUM X(16)
cardAccountId	NUMBER	Yes	11 digits, AIX	Owning account id	CARD-ACCT-ID 9(11)
cvvCode	NUMBER	Yes	3 digits (PCI)	Card verification value	CARD-CVV-CD 9(3)
embossedName	STRING	Yes	max 50	Name embossed on card	CARD-EMBOSSED-NAME X(50)
expirationDate	DATE	Yes		Card expiration date	CARD-EXPIRAION-DATE X(10) *(legacy typo)*
activeStatus	ENUM	Yes	CardStatus	Card active status	CARD-ACTIVE-STATUS X(1)

{{TODO: cardNumber and cvvCode are PCI-sensitive — define tokenization/masking strategy.}}

Used by: Card List, Card View, Card Update, Card Security, Authorization.

CardXref

Replaces CARD-XREF-RECORD (XREFFILE + CXACAIX / CVACT03Y, 50 bytes). Links card ↔ customer ↔ account; alternate index on accountId.

Attribute	Type	Required	Constraints	Description	Legacy Source
cardNumber	STRING	Yes	16 chars, key	Card number	XREF-CARD-NUM X(16)
customerId	NUMBER	Yes	9 digits	Customer id	XREF-CUST-ID 9(9)
accountId	NUMBER	Yes	11 digits, AIX	Account id	XREF-ACCT-ID 9(11)

Used by: Account View, Card List, Transaction Add, Authorization.

Transaction

Replaces TRAN-RECORD (TRANFILE / CVTRA05Y, 350 bytes). Key: transactionId.

Attribute	Type	Required	Constraints	Description	Legacy Source
transactionId	STRING	Yes	16 chars, unique key	Transaction identifier	TRAN-ID X(16)

Attribute	Type	Required	Constraints	Description	Legacy Source
typeCode	ENUM	Yes	TransactionType	Transaction type code	`TRAN-TYPE-CD X(2)`
categoryCode	ENUM	Yes	TransactionCategory	Transaction category code	`TRAN-CAT-CD 9(4)`
source	STRING	No	max 10	Transaction source	`TRAN-SOURCE X(10)`
description	STRING	No	max 100	Transaction description	`TRAN-DESC X(100)`
amount	NUMBER	Yes	decimal(11,2), signed	Transaction amount	`TRAN-AMT S9(9)V99`
merchantId	NUMBER	No	9 digits	Merchant id	`TRAN-MERCHANT-ID 9(9)`
merchantName	STRING	No	max 50	Merchant name	`TRAN-MERCHANT-NAME X(50)`
merchantCity	STRING	No	max 50	Merchant city	`TRAN-MERCHANT-CITY X(50)`
merchantZip	STRING	No	max 10	Merchant ZIP	`TRAN-MERCHANT-ZIP X(10)`
cardNumber	STRING	Yes	16 chars	Card used	`TRAN-CARD-NUM X(16)`
originTimestamp	DATE	No	timestamp	Origination timestamp	`TRAN-ORIG-TS X(26)`
processTimestamp	DATE	No	timestamp	Processing timestamp	`TRAN-PROC-TS X(26)`

Used by: Transaction List, Transaction View, Transaction Add, Transaction Posting, Reports.

TransactionCategoryBalance

Replaces TRAN-CAT-BAL-RECORD (`TCATBALF` / `CVTRA01Y`, 50 bytes). Composite key: `accountId` + `typeCode` + `categoryCode`.

Attribute	Type	Required	Constraints	Description	Legacy Source
accountId	NUMBER	Yes	11 digits, key part	Account id	`TRANCAT-ACCT-ID 9(11)`
typeCode	ENUM	Yes	TransactionType, key part	Transaction type	`TRANCAT-TYPE-CD X(2)`
categoryCode	ENUM	Yes	TransactionCategory, key part	Category	`TRANCAT-CD 9(4)`
balance	NUMBER	Yes	decimal(11,2), signed	Category running balance	`TRAN-CAT-BAL S9(9)V99`

Used by: Transaction Posting, Interest Calculation, Reports.

SecurityUser

Replaces SEC-USER-DATA (`USRSEC` / `CSUSR01Y`). Key: `userId`. Target FLOWX model gates access with roles; plaintext password is replaced by IdP credentials.

Attribute	Type	Required	Constraints	Description	Legacy Source
userId	STRING	Yes	8 chars, unique key	User id	`SEC-USR-ID X(8)`
firstName	STRING	Yes	max 20	First name	`SEC-USR-FNAME X(20)`
lastName	STRING	Yes	max 20	Last name	`SEC-USR-LNAME X(20)`
password	STRING	—	**DO NOT MIGRATE**	Plaintext password (legacy debt)	`SEC-USR-PWD X(8)`
userType	ENUM	Yes	UserType	Admin (`A`) / User (`U`)	`SEC-USR-TYPE X(1)`

`{{DECISION: Do NOT migrate the plaintext password column. Provision users in a modern IdP and map userType → FLOWX roles. Confirm IdP (Keycloak/other).}}`

Used by: Signon, User Administration.

SessionContext

Internal process data replacing the navigation portion of `CARDDEMO-COMMAREA` (`COCOM01Y`). Not persisted — lives in process instance data. Documented here for traceability; in FLOWX most of these become implicit process state.

Attribute	Type	Description	Legacy Source
userId	STRING	Signed-on user id	`CDEMO-USER-ID X(8)`
userType	ENUM	UserType	`CDEMO-USER-TYPE X(1)` (88 A/U)
selectedCustomerId	NUMBER	Current customer context	`CDEMO-CUST-ID 9(9)`
selectedAccountId	NUMBER	Current account context	`CDEMO-ACCT-ID 9(11)`
selectedCardNumber	STRING	Current card context	`CDEMO-CARD-NUM 9(16)`

`{{DECISION: The from/to tranid/program and last-map/mapset fields (CDEMO-FROM-*, CDEMO-TO-*, CDEMO-LAST-MAP) are pure navigation plumbing — replace with FLOWX process transitions and do not model them as data. Confirm.}}`

See Data Model Diagram for relationships.

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Candidate I/O contract

What a migrated implementation must produce so the scorer can compare it with the oracle. The contract fixes **formats only**. It says nothing about behaviour, so it may be handed to a builder without leaking the answers.

Inputs (given per run)

The same files the COBOL chain reads, one record per line, fixed width, layouts as in the copybooks:

File	Copybook	Record
`dailytran.dat`	CVTRA06Y	350
`acctdata.dat`	CVACT01Y	300
`cardxref.dat`	CVACT03Y	50
`tcatbal.dat`	CVTRA01Y	50
`discgrp.dat`	CVTRA02Y	50
`custdata.dat`, `carddata.dat`, `trantype.dat`, `trancatg.dat`	CUSTREC, CVACT02Y, CVTRA03Y, CVTRA04Y	500, 150, 60, 60

Run parameters: business date `2022-07-18`, INTCALC parameter `2022071800`.

Step isolation. A candidate for INTCALC receives the oracle's POSTTRAN outputs as its account and category-balance inputs, so an INTCALC result is never penalised for a POSTTRAN error.

Outputs

One JSON file per output: a list of objects, in any order. Field names and types are fixed. Strings may omit trailing spaces. Amounts are JSON strings or numbers holding at most two decimals (`"14.56"`, `-500`).

File	Step	Fields
`posttran_transact.json`	POSTTRAN	transaction fields (below) for every posted transaction
`posttran_dalyrejs.json`	POSTTRAN	transaction fields + `reason` (int) + `reason_desc` for every rejected transaction
`posttran_acctdata.json`	POSTTRAN	account fields for every account, after posting
`posttran_tcatbal.json`	POSTTRAN	`acct_id, type_cd, cat_cd, bal` for every category balance, after posting
`intcalc_systran.json`	INTCALC	transaction fields for every interest transaction written
`intcalc_acctdata.json`	INTCALC	account fields for every account, after interest

- Transaction fields: `tran_id, type_cd, cat_cd (int), source, desc, amt, merchant_id (int), merchant_name, merchant_city, merchant_zip, card_num, orig_ts, proc_ts`

- Account fields: `acct_id (int), active_status, curr_bal, credit_limit, cash_credit_limit, open_date, expiration_date, reissue_date, curr_cyc_credit, curr_cyc_debit, addr_zip, group_id`

A run may instead leave the oracle's fixed-width dumps (`<name>.txt`). The scorer reads either form.

Optional `flags.json`: the uncertainties the pipeline declared **before** scoring, as `[{"rule": "R-INT-TRUNC", "source": "<artefact path and line>"}]` or `[{"output": "...", "key": "...", "source": "..."}]`. See PROTOCOL.md §6 for what counts as a flag.